Journal of Environmental Studies and Sciences (2024) 14:400–414
https://doi.org/10.1007/s13412-024-00888-3

REVIEW ARTICLE

Revitalizing indigenous practices employed by farmers to reduce agriculture's vulnerability to climate change: a systematic review

Yohannes Yona Legide[1] · Getachew Sime Feyissa[2] · Tafesse Mathewos Karo[3]

Accepted: 29 December 2023 / Published online: 20 January 2024
© AESS 2024

Abstract
The purpose of this literature review is to rejuvenate the traditional climate change adaptation strategies employed by smallholder farmers in Sub-Saharan Africa. The study adopts a systematic approach, analyzing 77 relevant articles published in English between 2000 and 2022. The findings highlight the detrimental impact of climate change on African agriculture, which serves as the backbone of the continent's population. In response to the effects of climate change, smallholder farmers and agro-pastoral communities have devised and put into practice various indigenous adaptive strategies. These strategies include livelihood diversification, integrating livestock with crop production, incorporating agroforestry into farming systems, diversifying cropping patterns, adjusting planting schedules based on traditional weather forecasting, and conserving soil moisture through techniques like mulching and physical structures. These indigenous strategies have proven to be effective, affordable, sustainable, and easily adaptable, significantly bolstering communities' resilience to climate change-related disruptions. Nevertheless, the sustainable adoption of indigenous knowledge-based climate change adaptation strategies developed by smallholder farmers in Sub-Saharan Africa faces several challenges. These include insufficient documentation, limited support from younger and educated individuals, and inadequate integration with modern science. These obstacles hinder the widespread implementation and effectiveness of these valuable indigenous practices in addressing climate change impacts.

Keywords Indigenous knowledge · Climate adaptation · Resilience · Adaptive capacity

General background

Climate change has emerged as a global challenge with significant implications for sustainable development, particularly in developing countries within Sub-Saharan Africa (SSA) (Abate 2016; Harvey & Singh 2017; Mugambiwa 2018). Numerous studies have highlighted the adverse effects of temperature and precipitation changes on the livelihood security of farmers in these developing nations (Yigezu Wendimu 2021; Altieri et al. 2015). Smallholder farmers residing in rural areas of SSA face heightened vulnerability to climate-related extremes such as drought and flooding due to their reliance on rain-fed agricultural systems (Ayanlade et al. 2018; Clay 2019; Dynamics & Berkes 2006; Matewos 2019a, b). The limited access to modern agricultural tools further hinders their adaptive capacity, compelling smallholder farmers to rely on local adaptation methods and strategies. Existing studies (e.g., Matewos 2019a; Pearce & Ford 2015; Tschakert & Dietrich 2018; Van Huynh et al. 2020) have suggested the implementation of mitigation and adaptation strategies to mitigate the negative effects of climate change on smallholder farmers.

Mitigation strategies reduce the emission of greenhouse gasses, whereas adaptation-induced strategies entail actions that enable individuals and communities to cope with actual or potential climate change-induced impacts (Tschakert & Dietrich 2018). It is also argued that the rural smallholder

✉ Yohannes Yona Legide
johnyona99@yahoo.com

Getachew Sime Feyissa
abibiag@yahoo.com

Tafesse Mathewos Karo
tafessemk@gmail.com

[1] IPDR, Hawassa University, Hawassa, Ethiopia

[2] Department of Biology, Hawassa University, Hawassa, Ethiopia

[3] Department of Geography and Environmental Studies, Hawassa University, Hawassa, Ethiopia

farmers in SSA employ unique adaptation strategies that suit their socioeconomic, technological, and ecological contexts (Tschakert & Dietrich 2018). These strategies have been developed and employed by indigenous communities over many years to cope with climate-related and other shocks (Ayanlade et al. 2018; Rahut et al. 2021). Moreover, these context-specific strategies have played a vital role in ensuring the survival of communities and their agricultural systems in Sub-Saharan Africa (SSA) despite challenging environmental conditions and the impacts of climate change (Altieri et al. 2015; Magni 2017; Makate 2020; Shiferaw et al. 2013). The observation of this situation has led numerous scholars, such as Berkes and Jolly (2001), Abate (2016), Khatri et al. (2021), and Robison (2020), to raise doubts about the efficacy of conventional adaptation strategies in effectively reducing the vulnerability of rural livelihoods in a sustainable manner. Consequently, they advocate for the incorporation of indigenous adaptation strategies as an integral part of the overall approach. The indigenous adaptation strategies employed by smallholder farmers in Sub-Saharan Africa (SSA) are firmly grounded in their indigenous knowledge (IK) system.

IK refers to cumulative knowledge, beliefs, and a way of doing things that are passed down from one generation to another through proverbs, riddles, folktales, songs, legends and myths, culture, religion, stories, informal education, communication, and tradition (Abate and Kronk 2013; Obi et al. 2021). IK shapes farmers' interactions with their surroundings and evolves through practical experimentation of the knowledge-building process (Berkes 2012). Indigenous peoples share intergenerational, ancestry, and cultural aspects with a specific region in the face of environmental extremes, including climate change and variability (Crop et al. 2012; Makate 2020; Sharifian et al. 2022). This is the reason why IK is conceptualized as a location-specific lifelong learning system innovated and inherited by rural farmers over generations (Šūmane et al. 2018; Yua et al. 2022). Thus, indigenous climate change adaptation systems are considered an effective strategy for sustainable climate adaptation if they are integrated with modern scientific strategies (Jessen et al. 2022; Chanza & De Wit 2016; Adger et al., 2014; Songok et al., 2011). Likewise, Abate and Kronk (2013) argue that unless the community's IK and practices serve as a fulcrum for modern science-based techniques, the complex problems that farmers face due to climate change cannot be solved. This premise supports the argument for the complementarity of indigenous and scientific knowledge for sustainable climate change adaptation among smallholder rural farmers (Berkes & Jolly 2001). Likewise, Chanza and Wit (2013) advocated that amalgamation of modern agricultural science with IK could result in effective and sustainable climate change adaptation, which contributes to the present as well as future resilience of the community.

Recently, the international community has increasingly recognized the importance of indigenous knowledge systems (IKS) to shape climate change adaptation by enhancing the resilience of the local community (Abate 2016; Balehegn et al. 2019; Bussa et al. 2020; Mane et al., 2018; Yua et al. 2022). Accordingly, the United Nations Framework Convention on Climate Change (UNFCCC) and the Intergovernmental Panel on Climate Change (IPCC) both acknowledge the importance of IK to climate adaptation (Adger et al. 2014; Chanza and de Wit 2016; UNFCCC 2013) at the global level. IFAD (2016) also recognized the advantage of rural communities' indigenous innovations in improving their adaptive capacity to resist the adverse impacts of climate change and called for integration into regional and national development policies and strategies. Despite the increasing recognition of the importance of indigenous knowledge systems (IKS) in climate change adaptation by the global research community, the integration of IKS with formal Western technologically-based knowledge is still in its early stages and requires further attention (Chanza & Wit 2013). Additionally, empirical research findings that explore the interconnections between indigenous and scientific climate change adaptive strategies, particularly in Sub-Saharan Africa (SSA), are scattered and often not easily accessible to policymakers and practitioners (Asmamaw et al. 2020; Dynamics & Berkes 2006; Yua et al. 2022). Hence, the purpose of this literature review is to consolidate the key indigenous knowledge (IK) practices that have been formulated and adopted by smallholder farmers in Africa as a response to the detrimental impacts of climate change. The review primarily aims to identify various forms of knowledge, areas of application, and concerns related to the sustainable integration of IK with specific climate adaptation knowledge. The review is structured around the following central themes: the contextual background of indigenous climate change adaptation practices, the diverse domains of IK application, the cultural embeddedness of knowledge, and the challenges associated with co-producing knowledge.

Methodology/design/approach

Systematic literature retrieval and processing strategy

This review utilizes the IPCC (2014) framework of climate change adaptation, focusing on the intersection of indigenous knowledge (IK) and climate change adaptation. The analysis is based on a comprehensive examination of relevant literature, exploring the concepts of exposure, vulnerability, and adaptive capacity. Specifically, the term "vulnerability" refers to the system's ability to withstand the adverse impacts of climate change, including its susceptibility to

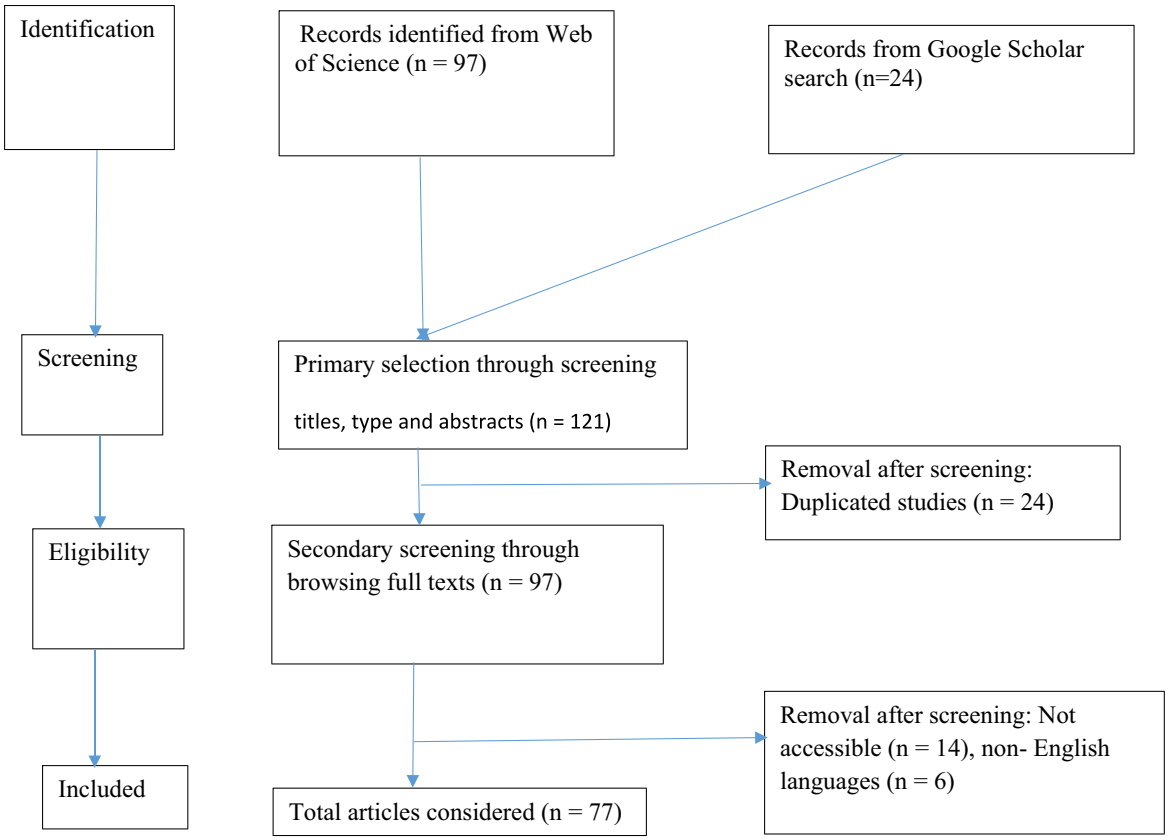


Fig. 1 PRISMA flow diagram showing review process; adopted from Khalid et al. (2015)

extreme events and unpredictability. The nature and severity of important climate changes encountered by a system are referred to as its exposure. Adaptive capacity, on the other hand, refers to a system's ability to modify its behaviors and expand its repertoire of coping mechanisms in response to current or predicted climatic change. In the real world, smallholder farmers adopt and use a wide range of indigenous adaptation approaches to reduce the likelihood and/or severity of poor outcomes caused by climate-related hazards. These tactics are intended to adapt to changing risks and pressures, with the goal of minimizing the negative consequences of such hazards.

By utilizing the employed framework, the authors able to identify indigenous adaptation strategies implemented by farmers. These strategies serve the purpose of reducing vulnerability and strengthening their capacity to cope with the consequences of climate change. To gather relevant information, we searched electronic databases, namely, Web of Science and Google Scholar, with a primary focus on peer-reviewed literature. Our search process adhered to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) protocol, as outlined by Khalid et al. (2015). The PRISMA protocol is critical in systematic reviews because it provides a consistent framework for performing, reporting, and evaluating research quality. Its use improves the overall rigor and transparency of systematic reviews and allows for evidence-based decision-making in a variety of disciplines of study.

We utilized phrases like "indigenous knowledge" and "climate change adaptation" to search the literature, even though the literature on the issues of IK and adaptation to climate change uses a variety of terminology. The term "Africa" was also included in the search criteria. The database searches were completed on June 5, 2022, and were limited to 2000–May 1, 2022. As a result, 121 documents were created, and their abstracts were assessed in light of the research topic. Additionally, free full-text articles with an emphasis on African nations that were written in English were filtered for further study. Following this, 44 articles that could not meet the inclusion criteria were rejected. After deleting duplicates from the two databases and screening for relevance, a total of 77 articles were chosen for further investigation (Fig. 1).

We used thematic-content analysis technique to conduct this systematic literature review. In order to evaluate the state of knowledge, this method helps to pinpoint the major themes and theoretical stances that are present throughout the text (Braun & Clarke, 2012). The data

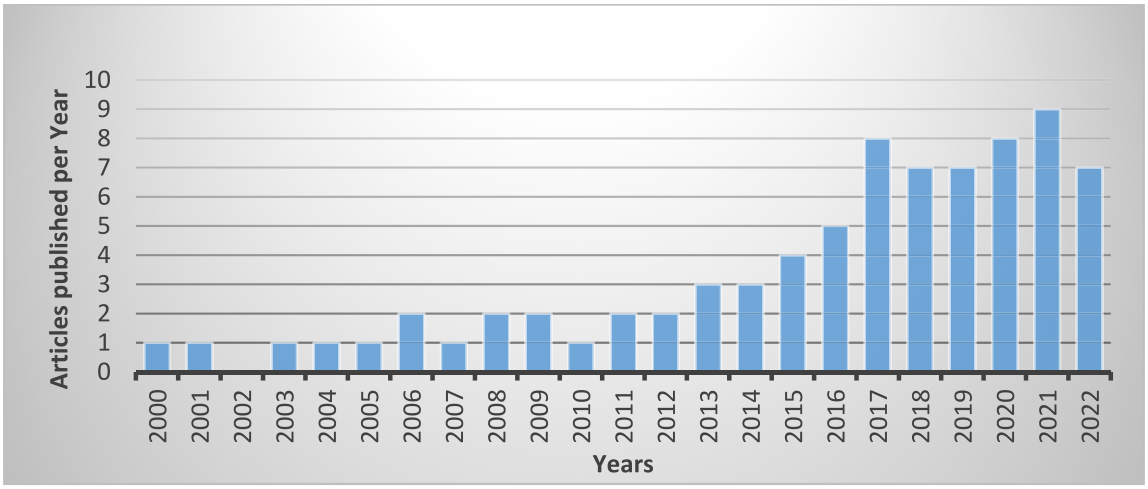
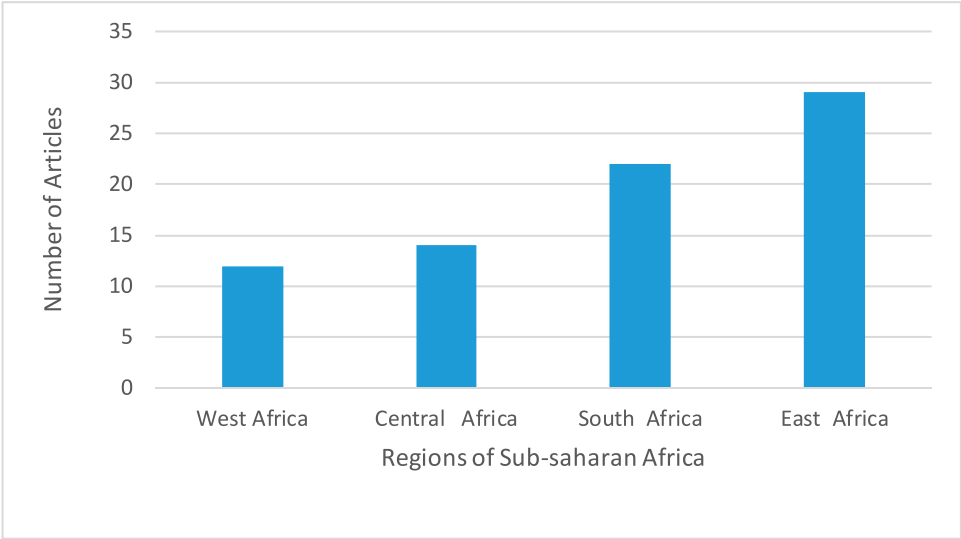


Fig. 2 Temporal distribution of reviewed articles

Fig. 3 Spatial distribution of reviewed articles



was collected, synthesized, and presented using specific methodologies. The initial step of the study team involved reviewing the abstracts, findings, and discussion sections of the selected publications to identify commonalities. Consequently, the authors evaluated indigenous adaptation strategies and organized them into 21 subthemes. In the subsequent phase, these subthemes were grouped into four categories based on their primary objectives. Lastly, the concepts were carefully examined to determine their significance, resulting in the identification of key indigenous knowledge-based climate change adaptation strategies employed by farming communities in Sub-Saharan Africa (SSA). These strategies included livelihood diversification, soil and water conservation, agroforestry, as well as weather and climate forecasting as depicted in Fig. 4.

Results and discussion

Introducing the literature

The majority of the 77 papers that were chosen were published in 2021, with 2017 and 2020 coming in second and third (Fig. 2). However, no literature for the year 2002 could be found in the articles that were reviewed.

Figure 3 illustrates how the different parts of SSA were addressed by the reviewed papers. The papers under review received the most contributions from East Africa, which contributed 38% ($n = 29$), South Africa, which contributed 29% ($n = 22$), and Central Africa, which contributed 18% ($n = 14$). The region with the lowest origination

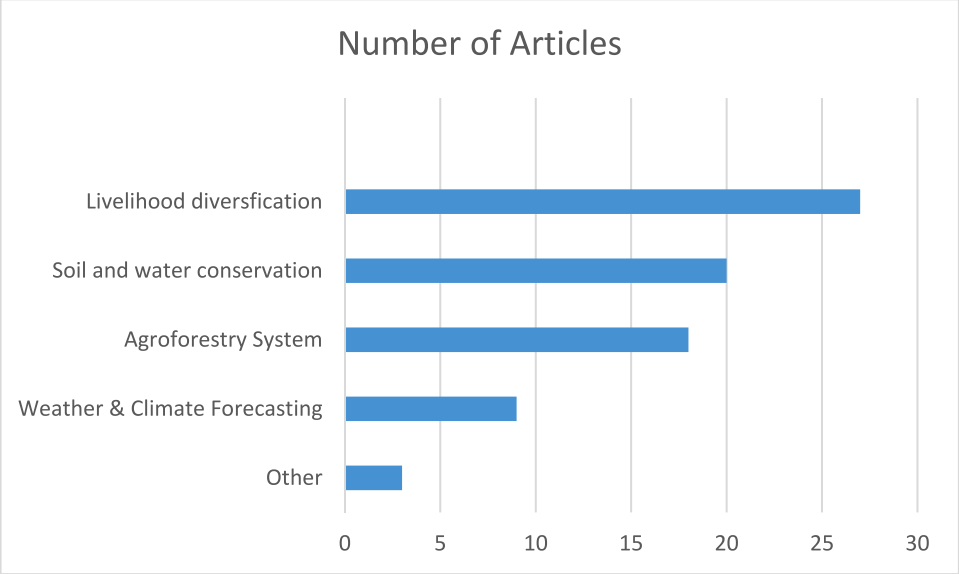


Fig. 4 Number of articles showing strategies adopted by farmers

rate was West Africa, contributed about 16% ($n = 11$) of publications (Fig. 3).

Value and context for the application of indigenous knowledge

This section merely sought to draw attention to a few of the indigenous adaptation techniques employed by smallholder farmers and the degree to which these techniques are essential for mitigating their vulnerability to climate-related impacts. We discovered from the review that the agricultural community in various parts of SSA has developed and adopted various types of IK and practices to mitigate the vulnerability of agriculture to climate change-related stresses (Fig. 4). Out of the reviewed articles, 35% dealt with livelihood diversification, followed by 25% that were associated with soil and water conservation-related knowledge. Likewise, agroforestry and climate early warning systems were discussed in 23% and 12% of the articles, respectively. The remaining 4% of the reviewed articles addressed a range of adaptation strategies, including migration and mobility.

Farmers developed the aforementioned indigenous adaptation practices to cope with drought, flooding, and rainfall variability (Maru et al. 2019; Ogalleh et al. 2012; Rahut et al. 2021; Singh 2012). Furthermore, smallholder farmers used their IK and practical skills for weather forecasting and vulnerability assessment (Bohensky & Maru 2011; Chanza & De Wit 2016; Maru et al. 2019; Shiferaw et al. 2013). Most of the IK-based vulnerability assessment findings and weather forecasting results were relatively consistent with those of scientific information disseminated through formal public institutions (Oliveira Andreotti et al. 2017). This argument sheds light on the current science-based climate change adaptation discourse dominated by a top-down orientation that fails to consider specific agroecological and socioeconomic contexts to integrate practice-based farmers' knowledge (Altieri et al. 2015; Mane et al., 2018; Yua et al. 2022).

Various studies revealed that African farmers developed and adopted multifaceted adaptation strategies for the problems they have faced due to climate change (Berkes & Jolly 2001). They developed a solution for every problem based on their long-lasting experimentation and experience (Briggs 2005; Mane et al., 2018). Some of the strategies targeted to solve the problem of food security (Chanza & Musakwa 2022); climate governance (Mugambiwa 2021); biodiversity conservation (Maru et al. 2019); and genetic resource conservation (Abate 2016; Altieri et al. 2015; Ayal et al. 2015). The reviewed literature posited that the development and adoption of IK-based adaptation strategies by these farmers contributed to the survival of the African agro-pastoralism system as a crucial livelihood system on the continents regardless of the prevailing harsh climatic conditions (Ayal et al. 2015; Debela 2017; Mugambiwa 2018, 2021; Rahman 2020). This is because the knowledge systems and associated behaviors increased the community's capacity for adaptation while simultaneously reducing the system's exposure and sensitivity to adverse climate-induced impacts (Clay, 2019; Abate 2016; Mugambiwa 2021; Mane et al., 2018).

African farmers responded to climate-induced food insecurity by adopting new ways in indigenous food production, processing, preservation, and food-sharing networks (Mabhaudhi et al. 2019; Shingirai S. Mugambiwa 2021; Senbeta et al. 2022; Sharifian et al. 2022). Based on their experience of recurring shortages of rainfall, the farmers in various countries of Africa intercropped various crops with different characteristics to solve the problem of yield losses and associated food insecurity. For instance, farmers

in Tanzania, Ghana, and Kenya intercropped highly productive and drought-resistant crops such as sweet potato, sorghum, and cassava with maize (Danso-Abeam et al., 2018; Mwaniki 2016; Theodory 2016). In the same vein, farmers in moisture-stress areas of Ethiopia also adopted enset, millet, and sorghum (Wondim et al. 2018; Rahut et al. 2021; Regassa & Stoecker 2012). Additionally, Chanza and Musakwa (2022), Berkes and Jolly (2001), Jessen et al. (2022), and Liehr et al. (2017) also reported the contribution of indigenous responses to adverse climate change, such as food insecurity. Using early-maturing crop types, integrating crops and livestock, and involving some of the members of their household in off-farm or non-farm activities all contributed to increasing adaptive capacity.

While farmers in different countries implemented country-specific indigenous adaptation strategies, there were also shared practices across countries that scaled at the continental level (Brondízio et al. 2021; Chikaire et al. 2018; de Oliveira Andreotti et al. 2017). In line with this, Magni (2017), Makate (2020), Gebrehiwot and Van Der Veen (2013), and IFAD (2016) comprehensively summarized indigenous knowledge (IK) practices adopted by rural communities in Africa. These practices include (a) resource diversification; (b) changes in crop varieties and species used; (c) adjustments in the timing of agricultural activities; (d) modifications in farming techniques; (e) shifts in location or mobility; (f) alterations in resources and/or lifestyles; and (g) sharing of resources with other groups. The purpose of this section is to highlight the primary indigenous adaptation strategies employed by smallholder farmers. Therefore, the following subsection provides a detailed explanation of these strategies.

Livelihood diversification-related indigenous knowledge

Diversification involves intentionally combining various activities to mitigate the risk of failure resulting from climate change and variability (Yua et al. 2022). This strategy has been employed by rural populations in Sub-Saharan Africa (SSA) for a long time as an adaptation technique to combat climatic variability and change (Maru et al. 2019; Theodory 2016; Upreti 1999). According to the literature, smallholder farmers have been observed to adopt diversification strategies in their income sources, crop selection, livestock herds, and migration routes as a means to reduce vulnerability to the impacts of climate change (Graham et al. 2013; Rahut et al. 2021; Sharifian et al. 2022). Additionally, Wondim et al. (2018) found that 30% of households in Gondar, Northern Ethiopia, generate income from off-farm and non-farm activities such as weaving, carpentry, poetry, and petty trade. In Burkina Faso, Reardon et al. (2001) identified a strong correlation between households' diversification into non-farm activities and their ability to withstand drought shocks.

Smallholder farmers purposefully allocated their land to cultivate a variety of crops (Van Huynh et al. 2020; Yua et al. 2022). Adger (2003) stated that diversifying food crops played a crucial role in reducing households' vulnerability to food shortages caused by crop failure resulting from climate change. Likewise, Xu et al. (2018) emphasized the significance of agro-fisheries in promoting diversified livelihoods and enhancing food security within communities in Benin. Overall, diversifying food crops provides more stability, resilience, and chances for sustainable livelihoods to smallholder farmers. Furthermore, IFAD (2016) echoed that indigenous livelihood diversification is an important strategy to reduce the risk of agricultural income due to drought. Traditionally, farming communities in SSA deliberately combined crop varieties with different water requirement statuses, such as water-loving and water-saving crops, early maturing and late maturing, under the philosophy of intercropping/mixed cropping (Bohensky & Maru 2011; Maru et al. 2019; Tigist Petros 2010). Intercropping refers to the practice of cultivating two or more crops within the same field (Upreti 1999). This agricultural technique is widespread in several countries across Sub-Saharan Africa (SSA), including Tanzania, Senegal, Ethiopia, Nigeria, and Burkina Faso. Intercropping practices in these regions involve various combinations, such as cultivating maize with common bean, maize with sorghum, coffee with enset, and incorporating multipurpose shed trees (Shiferaw et al. 2013; Tesfaye Abebe 2005; Yemataw et al. 2017; Brondízio et al. 2021; Bussa et al. 2020; Maru et al. 2019). These intercropping systems provide smallholder farmers with benefits such as taking advantage of different maturity periods, enhancing drought tolerance, and increasing overall yield (Tesfaye et al. 2014; Yemataw et al. 2017). Additional research conducted in Ethiopia (Chanza & Musakwa 2022; Dornan et al. 2014; Matewos 2020; Mekonnen et al. 2021; Rahut et al. 2021) has further validated the beneficial impact of traditional mixed cropping systems in bolstering the adaptive capacity of smallholder farmers. These studies demonstrate that indigenous knowledge (IK) and practices related to modifying cropping patterns play a significant role in enhancing smallholder farmers' ability to adapt to the impacts of climate change.

Migration is often regarded as a form of climate change adaptation strategy employed by individuals, households, and communities to mitigate the risks associated with the phenomenon. People, households, and communities utilize migration as a means to reduce the vulnerabilities stemming from climate change (Alhassan, 2019). Multiple studies (Balehegn et al. 2019; Koubi 2019; Yona & Mathewos 2017; Vinke et al. 2022) reviewed in this context highlight migration as an indigenous adaptation strategy implemented

to address environmental crises and diminish their impacts. In the context of moisture-stressed regions in Nigeria, Akinnagbe and Irohibe (2014) identified labor migration as a crucial climate change adaptation strategy adopted by smallholder farmers. Similarly, Theodory (2016) discovered that migration assumes growing significance as a component of rural livelihood strategies in Tanzania, particularly in response to the gradual impacts of climate change such as desertification, soil degradation, unpredictable rainfall patterns, and temperature fluctuations. Remittances served as supplementary income sources enabling migrants to better cope with the adverse impacts of climate change. The literature indicates that pastoralists in Sub-Saharan Africa (SSA) possessed a deep understanding of the necessity to diversify their herds by incorporating a combination of camels, cattle, and goats, thereby minimizing losses caused by recurring droughts (Balehegn et al. 2019; Muluken, 2020; Rahut et al. 2021).

Soil and water conservation-related knowledge

The productivity of agricultural activities and livelihoods reliant on agriculture is influenced by factors like soil fertility and water availability (Crop et al. 2018; Jessen et al., 2022). In response to this challenge, rural farmers have devised and adopted various indigenous strategies for soil and water management. These include practices such as contour plowing, mulching, and the utilization of trees for land reclamation (Wondim et al. 2018; Abate 2016; Maru et al. 2019).

The literature highlights that farmers in Sub-Saharan Africa (SSA) have a preference for locally developed physical structures and methods to combat soil erosion (Rahut et al. 2021). This inclination can be attributed, in part, to the failure of many modern soil and water conservation facilities due to inadequate design, construction, or maintenance. However, a significant number of farmers who employed indigenous physical soil and water conservation techniques, such as contour plowing, mulching, and tree-based land reclamation, found that their innovations were cost-effective, suitable for their agricultural systems, and easy to maintain (Corbeels et al. 2000; Rahut et al. 2021; Solomon et al., 2020). To enhance soil fertility and mitigate yield losses associated with climate change, Nigerian farmers have embraced various practices, including mulching, zero-tillage, agroforestry, land fallowing, and manure composting, as reported by Chikaire et al. (2018). Similarly, Wondim et al. (2018) documented that rural farmers in the northern region of Ethiopia employ agroforestry practices, intercropping, organic fertilizers (such as animal dung and crop residues), zero-tillage, and mulching. The authors further emphasized that these strategies have effectively addressed the issue of soil erosion within the community.

Agroforestry system-related indigenous knowledge

Agroforestry entails integrating crop-based agricultural systems with livestock farming in a close symbiotic relationship rather than simply combining multipurpose trees with edible fruits, where the crop sector provided animal feed and the livestock sector provided fertilizer and traction power (Bohensky & Maru 2011; Maru et al. 2019; Tesfaye Abebe 2005). Various agroecosystem farming strategies, such as agroforestry, silvopastoral, and poly-cultural systems, have proved their ability to adapt to and ameliorate the effects of climate change (Gebrehiwot & Van Der Veen 2013; Maru et al. 2019; Schlingmann et al. 2021). The varied range of techniques provides numerous strategies for addressing climate change concerns and fostering the development of sustainable agriculture systems.

Research has demonstrated that agroforestry-based indigenous knowledge (IK) plays a vital role in natural resource management, promoting sustainable development, and enhancing agricultural production systems (Abate 2016; Briggs 2005; Brondízio et al. 2021; Mutambisi et al. 2021). Findings from studies conducted in South Africa and Ethiopia further support this argument (Mutambisi et al. 2021; Bohensky & Maru 2011; Maru et al. 2019; Wondim et al. 2018; Mugambiwa & Makhubele 2021). Mugambiwa and Makhubele (2021), Debela (2017), Debelo (2020), and Deressa et al. (2010) highlight those small-scale farmers, who face adverse impacts from climate change, have adopted the integration of livestock into crop production as a crucial indigenous strategy. The integration of livestock into crop production has fostered a harmonious socio-ecological system across the majority of Africa (Bussa et al. 2020; More and Madolo 2020). Home-gardens, enset-based agroforestry systems, cocoa, coffee, and banana farming are among the most prevalent indigenous farming systems in Africa, encompassing various distinct types of agroforestry systems (Brondízio et al. 2021; Mutambisi et al. 2021; Yemataw et al. 2017; Senbeta et al. 2022). Specifically, the enset-based agroforestry system in southern and southwest Ethiopia has played a significant role in supporting over 20 million people in achieving sustainable livelihoods and food security (Auge et al., 2018; Senbeta et al. 2022; Yemataw et al. 2017).

Similarly, More and Madolo (2020) highlighted the advantages of various local knowledge systems related to forests utilized by farmers in the Lemo district of the Hadiya zone in southern Ethiopia. This approach not only enhanced the adaptive capacity of individual farmers but also bolstered the resilience of the entire socio-ecological system in the region. The Gedeo people in southern Ethiopia were also well-versed in agroforestry practices centered on coffee production (Maru et al. 2019). This traditional agroecological knowledge played a crucial role in maintaining the resilience of their farming system despite challenges such as high

population density and escalating climatic stressors. Senbeta et al. (2022) further exemplified the benefits of an enset-based agricultural system in the densely populated highlands of southern and southwestern Ethiopia. This system emerged as a prominent adaptation strategy, enhancing the farming system's resilience to climate change. Given these positive outcomes, it is crucial to disseminate these effective techniques to other regions with similar agroecological conditions in order to reduce farmers' vulnerability to the impacts of climate change.

Indigenous knowledge and weather and climate forecasting

In countries across Sub-Saharan Africa (SSA) including Benin, South Africa, Nigeria, Kenya, and Ethiopia, rural communities have devised and put into practice climate change early warning systems that utilize local indigenous knowledge (IK) to anticipate and forecast drought-related events (Jessen et al., 2022; Mclean & Nakashima 2012; Muyambo & Thomas 2016). Muyambo & Thomas (2016) documented how smallholder farmers in the Eastern Cape Province of South Africa developed a sophisticated system for collecting, analyzing, and projecting temperature and rainfall data. This system enabled them to determine optimal cropping patterns and planting times. Additionally, these farmers employed various strategies within the livestock industry to cope with climate challenges. These included providing emergency fodder during drought periods, maintaining diverse livestock species to enhance resilience to climate extremes, adopting nomadic movement patterns, slaughtering weak livestock for food, and transitioning to more suitable cattle breeds (Hernandez et al. 2022; Jessen et al., 2022; Komba & Muchapondw 2015; Muyambo & Thomas 2016). Furthermore, pastoralist communities in Ethiopia's Afar region and Borana zone have developed a range of adaptation techniques to cope with the persistent drought in their areas (Balehegn et al. 2019; Debela 2017). Drawing from their extensive experience and experimentation, these communities have acquired and applied indigenous weather forecasting knowledge and skills. Building on their historical ability to predict weather patterns, these pastoralists make informed decisions regarding their mobility, herd diversity, livestock numbers, and the type of farming system they rely on for income. Scholars have recognized the value of traditional agroecological knowledge and the competence of these communities in adaptation decision-making processes (Lewis, 2017; Sharifian et al. 2022).

Smallholder farmers across various African regions employ diverse indicators to forecast climate variability and change, providing valuable insights for agricultural decision-making (de Oliveira Andreotti et al. 2017; Rahman 2020). Various studies conducted by Balehegn et al. (2019) in Ethiopia, Jessen et al. (2022) in Benin, Mugambiwa and Makhubele (2021) in Zimbabwe, Theodory (2016) in Tanzania, and Muyambo & Thomas (2016) in South Africa have all demonstrated that rural communities utilize their indigenous weather forecasting knowledge and skills to validate and verify scientific knowledge. De Oliveira Andreotti et al. (2017) further assert that small-scale farmers in numerous Sub-Saharan African countries rely on a range of indicators, including plant phenology, animal behavior, moon phases, and cloud movement, among others, to predict the timing, intensity, and distribution of rainfall. The authors also stated that farmers use their knowledge of weather trends to make informed judgments about crop selection and livestock management on their farms. Consequently, their weather knowledge and forecasting skills are a critical resource for enhancing their resilience (Briggs 2005; Muyambo & Thomas 2016; Sharifian et al. 2022).

In Africa, subsistence farmers frequently integrate their lifelong climatological knowledge and wisdom with climate information offered by formal government institutions (Corbeels et al. 2000; Crop et al. 2018). Rural communities often gravitate towards indigenous knowledge (IK) systems that are readily accessible and easy to adopt, as modern weather forecasting services are often perceived as too complex to understand and utilize effectively (Balehegn et al. 2019; Mafongoya, P. L., and Ajayi, 2017). The close bond between rural communities and their natural environment facilitated the observation, understanding, interpretation, and anticipation of local weather conditions. This demonstrates how diverse indigenous knowledge (IK) systems and practices employed by African communities enable them to make informed decisions that mitigate the adverse impacts of climate change and enhance their resilience (Balehegn et al. 2019; Mugambiwa 2018).

Afar and Borana pastoralists in Ethiopia effectively utilize indigenous knowledge (IK) for weather forecasting, along with their access to traditional meteorological data. This enables them to make informed decisions, such as relocating their herds to areas with available water and pasture or storing pasture for periods of drought, thus reducing their vulnerability to drought (Balehegn et al. 2019; Debela 2017). Additionally, Mclean and Nakashima (2012) argued that the indigenous weather forecasting system adopted in Eastern African countries helps address the gaps in formal seasonal forecasting conducted by public institutions, with the support of agricultural extension agents. Many African countries' formal climate information services have been criticized for their inaccessibility and lack of adaptation (Basdew et al. 2017). In contrast, indigenous weather services are easily accessible through indigenous social networks, making them a more accessible and user-friendly option (Balehegn et al. 2019; Rahman 2020). As a result, indigenous knowledge (IK) and related weather forecasting information

serve as valuable, adaptable, affordable, and sustainable community resources. These resources play a critical role in enhancing the capacity of knowledge holders to adapt to changing conditions (Chikaire et al. 2018; Xu et al. 2018). Additionally, Briggs (2005) and Jessen et al. (2022) have highlighted IK as an asset for marginalized social groups, enabling them to mitigate vulnerabilities and improve their ability to adapt to the impacts of climate change.

The need for integrated knowledge system

Knowledge co-production involves the active collaboration and integration of modern science-based knowledge and indigenous knowledge of climate change adaptation (Altieri et al. 2015; Bohensky & Maru 2011; Van Huynh et al. 2020; Yua et al. 2022). This process goes beyond simply recognizing and acknowledging the value of indigenous knowledge; it entails a co-creative and collaborative approach where different knowledge systems are combined, bridged, and mutually informed to generate new insights, understandings, and solutions for addressing climate change challenges. It emphasizes the equal participation and contribution of diverse knowledge holders, fostering a shared learning process that integrates different perspectives, approaches, and experiences. By merging these knowledge systems, knowledge co-production aims to enhance the effectiveness, relevance, and sustainability of climate change adaptation efforts (Van Huynh et al. 2020). Among the articles reviewed, only seven studies (Achparaki et al., 2012; Makate 2020; Reniko et al. 2018; Mane et al., 2018; Yua et al. 2022) specifically investigated the advantages of integrating both knowledge systems for achieving effective and sustainable climate change adaptation in Sub-Saharan Africa (SSA). Per findings from other authors, small-scale farmers have embraced a combination of traditional and new adaptation strategies to mitigate the adverse impacts of climate change (Chikaire et al. 2018; Magni 2017; Theodory 2016). Similarly, Bohensky and Maru (2011) and Chanza and De Wit (2016) emphasized that integrating modern science-based adaptation strategies with conventional practices yields greater benefits than deploying these approaches separately. By integrating both knowledge systems, additional advantages can be achieved, such as reducing the vulnerability of farm households to climate-related shocks.

In African rural communities, the integration of multiple activities, such as the crop-livestock system, is a common practice (Oliveira A. et al., 2017). Instead of implementing indigenous or scientific adaptation strategies in isolation, integrating both approaches can help address the issue of maladaptation (Basdew et al. 2017; Mane et al., 2018; Yua et al. 2022). By combining indigenous and scientific techniques, a more comprehensive and effective approach to climate change adaptation can be achieved. Moreover, it enhances households' confidence in utilizing their indigenous knowledge (Oliveira A. et al., 2017). Consequently, the integrated application of both knowledge systems is crucial in addressing the limitations associated with each system (Balehegn et al. 2019; Brondízio et al. 2021; Davoudi et al. 2012; Mane et al., 2018; Yua et al. 2022). Each knowledge system possesses its own inherent constraints. For example, the traditional understanding of weather forecasting among South African farmers has been criticized by Basdew et al. (2017) for its lack of predictability and reliability. The mentioned authors also expressed concerns about the existing science-based forecasting system, highlighting its limitations in terms of affordability, accessibility, and application to address climate change impacts. This indicates that each knowledge system has its own set of limitations when utilized independently. To overcome these limitations, a combination of the scientific weather forecasting system, which may have shortcomings, with the indigenous system, which is more widely available and cost-effective, could provide a potential solution (Wondim et al. 2018; Corbeels et al. 2000; Debela 2017; Maru et al. 2019; Mugambiwa & Makhubele 2021; Theodory 2016).

Reed et al. (2013) presented empirical evidence supporting the advantages of integrating local and scientific information, showcasing how this collaboration enables communities to develop reliable indicators for monitoring changes in a fast and efficient manner. However, for the successful synergization of these knowledge bases, it is essential to establish mutual respect between holders of indigenous knowledge and experts in modern knowledge. Without this prerequisite, the dichotomy persists, and rural populations are unable to benefit from sustainable adaptation measures (Asmamaw et al. 2020; Balehegn et al. 2019; Nyong et al. 2007).

To effectively address climate change, researchers must have a comprehensive understanding of the cultural traditions and information dissemination practices within the local communities. This highlights the integration of cultural values into both climate change adaptation strategies and the channels through which these strategies are communicated. For instance, certain indigenous knowledge systems in Ethiopia embody unique cultural values that facilitate effective communication, reducing information gaps and promoting equitable understanding between different stakeholders (Alem et al., 2018; Balehegn et al. 2019; More and Madolo 2020; Matewos 2019b). Moreover, indigenous knowledge systems play a crucial role in strengthening the capacity of local institutions for adaptation by mobilizing resources, fostering community participation, and disseminating reliable adaptation strategies (Chanza & De Wit 2016; Liehr et al. 2017; Mugambiwa 2018).

The embeddedness of IK in the community's culture and local institutions

Indigenous knowledge is deeply rooted in the culture and local institutions of communities (Kebede & Belay 2017; Makate 2020). Culture plays a vital role in the preservation and acceptance of indigenous knowledge and practices by establishing rules and procedures that govern the interactions between humans and nature (Chanza & Musakwa 2022). Community members align their behavior with socio-cultural norms when engaging with their environment. An example of the influence of cultural norms can be observed in Zimbabwe, where indigenous populations have benefited from the conservation of water bodies, wildlife, and vegetation through the use of "taboos" and social sanctions derived from indigenous institutions and traditional legal systems (Reniko et al. 2018; Yua et al. 2022). These communities have established rules and sanctions that regulate access to these resources, effectively controlling the overexploitation of valuable common-pool resources, including wetlands, community forests, and range lands, in various regions of Africa (Balehegn et al. 2019; Basdew et al. 2017; Debela 2017; Fadina & Barjolle 2018; Mwaniki 2016; Rahut et al. 2021; Theodory 2016). In agriculture-based livelihoods, the importance of indigenous institutions to climate change adaptation and livelihood enhancement across traditional farming systems in African nations was further emphasized in Institutions and Adaptation (2008) and Sumane et al. (2018). Furthermore, Mesfin et al. (2020) claimed that Ethiopian indigenous/informal/institutions (like iddir,[1] iqub,[2] and debo[3]) assisted in labor and financial mobilization during a difficult time. Such institutions helped to reduce the risk of crop and livestock damage due to climate-related hazards (Balehegn et al. 2019; Debela 2017; Maru et al. 2019). This implies that indigenous institutions contributes to reducing the vulnerability of the community to climate change-induced shocks (Aredo 2010).

Numerous studies in Africa, such as those by Kebede and Belay (2017), Makate (2020), Maru et al. (2019), and Matewos (2019a, b), demonstrated the importance of local institutions in aiding local adaptation strategies by offering weather forecasts, mediating information exchange, and managing resource governance. Similar research found that the IK of pastoralists in Ethiopia played a crucial role in managing the rangelands and water points required for climate change, according to Balehegn et al. (2019) and Debela (2017). This demonstrates how IK and related institutions are essential for boosting rural communities' socio-ecological resilience and promoting adaptation potential for climate change shocks. Understanding local institutions and cultural dynamics is therefore essential to developing sustainable strategies for adapting to climate change impacts (Aredo 2010; Matewos 2019b; Ochieng & Ahmed 2014; Šūmane et al. 2018).

Challenges in using indigenous adaptive knowledge

According to the literature review, IK and related adaption practices are increasingly lost their importance in the present climate change adaptation discourse irrespective of their contribution to adaptive resilience (Balehegn et al. 2019; Basdew et al. 2017; Mafongoya, P. L. and Ajayi, 2017; Magni 2017; Makate 2020; Robison 2020). The authors also noted that the limited adoption of indigenous climate and related adaptation strategies in Africa was mostly due to negative perceptions of indigenous knowledge and the deterioration of traditional cultures as a result of industrialization. For example, Mafongoya and Ajayi (2017) argued that people with formal education and the younger generation do not value indigenous knowledge. They added that once-sacred places and taboo were no longer held in the same esteem. In this context, Alem et al. (2018) noted that inadequate documentation was a driving factor for the declining application of indigenous knowledge as a strategy for adaptation. Similar findings were reported by Balehegn et al. (2019) and Debela (2017) on the limited attention paid by young people and educated individuals, such as agricultural extension workers. This condition exerted a threat to the extensive application of indigenous knowledge on climate change adaptation and its sustainability. Accordingly, Senbeta et al. (2022) identified the enset farming system's replacement with other annual crops as a threat to the indigenous farming system in Ethiopia.

What is missing?

Through our analysis, we have identified several research gaps that require greater scientific attention. Firstly, the majority of researchers focusing on pro-indigenous knowledge-based adaptation to climate change assert that implementing these strategies significantly contributes to achieving the three pillars of sustainable development: economic, social, and environmental sustainability (Berkes & Jolly 2001; Jessen et al. 2022; Mafongoya and Ajayi 2017; Sharifian et al. 2022). Nevertheless, the majority of the literature examined tends to place greater emphasis on the economic and environmental advantages of employing indigenous adaptation strategies. Specifically, there is

[1] Iddir is indigenous voluntary associations established primarily to provide mutual support to address community concerns.

[2] Iqub is an association of people having common objectives of mobilizing resources, especially finance, and distributing it to members on rotating basis.

[3] Debo is the organization of a community to help those in need of labor; it may be to help harvest a farm or renovate houses. The target groups are usually the elderly, widowers, or handicapped people.

heightened attention given to practices that contribute to economic sustainability, such as the integration of farming and livestock systems, livelihood diversification, intercropping, and migration. On the other hand, agroforestry, soil and water conservation techniques, water harvesting, crop rotation, manuring, and the construction of various physical structures are commonly associated with environmental sustainability. Secondly, there is a noticeable scarcity of studies that delve into the social aspect of adopting indigenous adaptation strategies. In particular, the literature lacks sufficient coverage of the gender dimension of indigenous climate change adaptation. It is worth noting that female farmers not only face greater vulnerability to climate-induced risks compared to their male counterparts due to the added burden of non-climate-related factors but they also play a crucial role as custodians of indigenous climate adaptation strategies. Despite this significance, the literature falls short in adequately addressing the gender-specific implications of indigenous climate change adaptation. Furthermore, it is critical to note that the implementation of climate change adaptation techniques does not occur independently of policy frameworks. The successful integration of indigenous and scientific adaptation solutions, in particular, necessitates significant national policy backing (Chanza & Wit 2013; Shikur 2020).

According to the majority of the studies being evaluated (Abate 2016; Balehegn et al. 2019; Brondízio et al. 2021; Yua et al. 2022), both knowledge systems must complement one another. However, none of the literature provided recommendations for how to create laws that would permit the blending of the two knowledge systems. In order to realize a resilient community that is resistant to all climate-related risks, we recommended that national agriculture policies as well as climate adaptation policies should give adequate opportunities for the integration of scientific knowledge with the community's IK. Furthermore, the reviewed literature does not adequately address the function of IK in climate adaptation and governance through indigenous institutions. There is a wealth of research on the function of local institutions in managing common-pool natural resources, coping with the effects of climate change, and addressing food insecurity. None of the evaluated literature mentioned the function of indigenous informal institutions in reducing vulnerability and boosting their adaptive potential, with the exception of Balehegn et al. (2019) and Debela (2017), who studied the social network of pastoralists in Ethiopia. The majority of the reviewed literature concentrated on the role of local government organizations, which are responsible for shaping the majority of the planned adaptation actions.

Summary, conclusion, and future directions

Based on the review, it has been observed that climate change hurts African agriculture, primarily through drought, high temperatures, disease, and pest infestation. To mitigate these effects, smallholder farmers have developed and embraced various indigenous knowledge (IK) and adaptive strategies. These strategies include weather forecasting, diversification of livelihoods, agroforestry, water harvesting, soil fertility management, crop management (such as intercropping, growing drought-resistant varieties, changing crop types and planting times), and labor migration. By implementing these adaptation strategies, smallholder farmers have enhanced their resilience against the adverse consequences of climate change. This is because IK not only contributes to increasing the resilience of communities but also helps reduce their vulnerability to negative impacts. Additionally, the use of practice-based techniques rooted in IK offers efficient resource utilization and cost-effectiveness.

It can be concluded that climate change not only poses a threat to the livelihoods of smallholder farmers but also presents an opportunity to embrace their local wisdom and practices. However, there are current challenges to the sustainable application and preservation of IK in climate adaptation. These challenges include insufficient documentation, a lack of attention from educated youth towards IK, and the erosion of cultural values associated with the knowledge system. Therefore, future development policies, particularly those related to climate adaptation, should prioritize the integration of IK and its associated adaptation practices alongside modern science-based approaches. Moreover, it is crucial to educate and raise awareness among young people and those in formal education systems about the importance of IK. This is essential to enhance the applicability and sustainability of IK systems in Sub-Saharan Africa. Consequently, incorporating IK into the educational curriculum can contribute to the decolonization of IK and the promotion of associated climate adaptation strategies.

Author contribution All of the three authors actively engaged in all aspects of research process starting from designing, methodology development, conceptualization, data curation, writing original draft, reviewing, and editing equally. Finally, all the authors have read and agreed to the published version of the manuscript and claim equal authorship.

Data Availability The authors confirm that the data supporting the findings of this study are available within the article.

Declarations

Competing interests The authors declare no competing interests.

References

Abebe T (2005) Diversity in home garden agroforestry systems in southern Ethiopia. In Tropical Resource Management: PhD Thesis, Wageningen University. Research@WUR; https://research.wur.nl/en/publications/

Abate T (2016) Contribution of indigenous knowledge to climate change and adaptation response in southern Ethiopia. J Earth Sci Clim Chang 7(11). https://doi.org/10.4172/2157-7617.1000377

Abate RS, Kronk EA (2013) Commonality among Unique Indigenous Communities: An Introduction to Climate Change and Its Impacts on Indigenous Peoples. Climate Change and Indigenous Peoples. The Search for Legal Remedies. Edward Elgar Publishing. https://ssrn.com/abstract=2213506

Achparaki M, Thessalonikeos E, Tsoukali H, Mastrogianni O, Zaggelidou E, Chatzinikolaou F, Vasilliades N, Raikos N (2012) Heavy metals toxicity. Aristotle University. Med J 39(1):20–31. https://doi.org/10.26226/morressier.5a9f5e0fd462b8028c7e7d3c

Adger WN, Brown K, Fairbrass J, Jordan A, Paavola J, Rosendo S, Seyfang G (2003) Governance for sustainability: Towards a 'thick' analysis of environmental decision-making. Environ Plan A 35(6):1095–1110. https://doi.org/10.1068/a35289

Adger WN, Pulhin JM, Barnett J, Dabelko GD, Hovelsrud GK, Levy M, Oswald Spring U, Vogel CH (2014) Human security. In: Climate change 2014: Impacts, adaptation, and vulnerability. Working Group II to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, pp 755–791

Akinnagbe OM, Irohibe IJ (2014) Agricultural adaptation strategies to climate change impacts in Africa: a review. https://www.researchgate.net/publication/272364792

Alem H, Lien G, Hardaker JB, Guttormsen A (2018) Regional differences in technical efficiency and technological gap of Norwegian dairy farms: A stochastic meta-frontier model. Appl Econ 51(4):114. https://doi.org/10.1080/00036846.2018.1502867

Alhassan H (2019) Farmers choice of adaptation strategies to climate change and variability in arid region of Ghana. Rev Agric Appl Econ 22(1):32–40. https://doi.org/10.15414/raae.2019.22.01.32-40

Altieri MA, Nicholls CI, Henao A, Lana MA (2015) Agroecology and the design of climate change-resilient farming systems. Agron Sustain Dev 35(3):869–890. https://doi.org/10.1007/s13593-015-0285-2

Aredo D (2010) The Iddir: An Informal Insurance Arrangement In Ethiopia: Savings and Development, Vol. 34, No. 1, pp. 53–72 Published by: Centre for Socio-economic Dynamics and Cooperation of the University of Bergamo Stable URL: https://www.jstor.org/stable/41406511

Asmamaw M, Mereta ST, Ambelu A (2020) The role of local knowledge in enhancing the resilience of Dinki watershed social-ecological system, central highlands of Ethiopia. PLoS ONE 15(9 September):1–18. https://doi.org/10.1371/journal.pone.0238460

Augé E, Pelegrí C, Manich G, Cabezón I, Guinovart JJ, Duran J, Vilaplana J (2018) Astrocytes and neurons produce distinct types of polyglucosan bodies in Lafora disease. Glia 66(10):2094–2107. https://doi.org/10.1002/glia.23463

Ayal DY, Desta S, Gebru G, Kinyangi J, Recha J, Radeny M (2015) Opportunities and challenges of indigenous biotic weather forecasting among the Borena herders of southern Ethiopia. SpringerPlus 4(1). https://doi.org/10.1186/s40064-015-1416-6

Ayanlade A, Radeny M, Akin-Onigbinde AI (2018) Climate variability/change and attitude to adaptation technologies: a pilot study among selected rural farmers' communities in Nigeria. GeoJ 83(2):319–331. https://doi.org/10.1007/s10708-017-9771-1

Balehegn M, Balehey S, Fu C, Liang W (2019) Indigenous weather and climate forecasting knowledge among Afar pastoralists of north eastern Ethiopia: role in adaptation to weather and climate variability. Pastoralism 9(1). https://doi.org/10.1186/s13570-019-0143-y

Basdew M, Jiri O, Mafongoya PL (2017) Integration of indigenous and scientific knowledge in climate adaptation in KwaZulu-Natal, South Africa. Chang Adapt Socio-Ecol Syst 3(1):56–67. https://doi.org/10.1515/cass-2017-0006

Berkes F (2012) Sacred ecology: traditional ecological knowledge and resource management. Taylor and Francis, London

Berkes F, Jolly D (2001) Adapting to climate change: social-ecological resilience in a Canadian western Arctic Community. Conserv Ecol 5(2):18. https://doi.org/10.1016/S0964-5691(01)00060-6

Bohensky EL, Maru Y (2011) Indigenous knowledge, science, and resilience: What have welearned from a decade of international literature on"integration"? Ecol Soc 16(4):6. https://doi.org/10.5751/ES-04342-160406

Braun V, Clarke V (2012) Thematic analysis. In: Cooper H, Camic PM, Long DL, Panter AT, Rindskopf D, Sher KJ (eds) APA handbook of research methods in psychology, vol 2

Briggs J (2005) The use of indigenous knowledge in development: problems and challenges. Prog Dev Stud 5(2):99–114. https://doi.org/10.1191/1464993405ps105oa

Brondízio ES, Aumeeruddy-Thomas Y, Bates P, Carino J, Fernández-Llamazares Á, Ferrari MF, Galvin K, Reyes-García V, McElwee P, Molnár Z, Samakov A, Shrestha UB (2021) Locally based, regionally manifested, and globally relevant: indigenous and local knowledge, values, and practices for nature. Annu Rev Environ Resour 46:481–509. https://doi.org/10.1146/annurev-environ-012220-012127

Bussa B, Babu A, Godana N, Universities H, Universities H (2020) Assessment of farmers' indigenous knowledge on soil conservation; a case of Karcha district; southern Ethiopia. Journal of Indigenous Knowledge 2:1–23. http://213.55.95.79/index.php/JIKDS/article/view/1010

Chanza N, De Wit A (2016) Enhancing climate governance through indigenous knowledge: case in sustainability science. S Afr J Sci 112(3–4):1–7. https://doi.org/10.17159/sajs.2016/20140286

Chanza N, Musakwa W (2022) Revitalizing indigenous ways of maintaining food security in a changing climate: review of the evidence base from Africa. Int J Clim Chang Strateg Manag 14(3):252–271. https://doi.org/10.1108/IJCCSM-06-2021-0065

Chanza N, Wit AD (2013) Epistemological and methodological framework for indigenous knowledge in climate science. Afr J Indigenous Knowl Syst 12(2):203–216

Chikaire JU, Ejiogu NEO, Emerhirhi E (2018) Sustainable indigenous knowledge systems in agriculture for climate change adaptation and mitigation in southeast Nigeria. Int J Curr Res Acad Rev 6(9):97–107. https://doi.org/10.20546/ijcrar.2018.609.010

Clay MM (2019) An observation survey of early literacy achievement, 4th edn. Heinemann, Portsmouth, NH

Corbeels M, Shiferaw A, Haile M (2000) Farmers' knowledge of soil fertility and local management strategies in Tigray. Ethiop Manag Afr Soils 10:30

Crop A, Society S, Tovignan S D (2012) Local knowledge and adaptation to climate change in Ouémé. Valley, Benin. Afr Crop Sci J 20:181–192

Crop A, Society S, Fungo R, Tieguhong JC, Muyonga JH, Odjo S, Tchingsabe O, Tchatat M, Box PO (2018) Perceived nutrition benefits and socio-demographic factors affecting consumption of forest foods in Eastern and Sub-Sahara African countries, including Cameroon, have high prevalence of undernutrition compared to recommended limits of 2 % for stunting a. Afr Crop Sci J 26(2):203–217. https://doi.org/10.4314/acsj.v26i2.4

Danso-Abbeam G, Ehiakpor DS, Aidoo R (2018) Agricultural extension and its effects on farm productivity and income: insight from Northern Ghana. Agric Food Secur 7(1):1–10. https://doi.org/10.1186/s40066-018-0225-x

Davoudi S, Shaw K, Haider LJ, Quinlan AE, Peterson D, Wilkinson C, Fünfgeld H, Mcevoy D, Davoudi S, Davoudi, S, Shaw K, Haider LJ, Quinlan AE (2012) Resilience: a bridging concept or a dead end? "Reframing" resilience: challenges for planning theory and practice interacting traps: resilience assessment of a pasture management system in northern Afghanistan urban resilience: what does it mean in. 9357(May):299–333. https://doi.org/10.1080/14649357.2012.677124

Debelo WB (2020) Review on Roles and Challenges of Agricultural Extension System on Growth of Agricultural Production in Ethiopia. Journal of Plant Sciences 8(6):189–200

de Oliveira Andreotti V, Mika C, Ahenakew C, Hireme H (2017) Indigenous knowledge systems and anticipation. In Handbook of anticipation (Issue November). https://doi.org/10.1007/978-3-319-31737-3_40-1

Deressa TT, Hassan RM, Ringler C (2010) Perception of and adaptation to climate change by farmers in the Nile Basin of Ethiopia. J Agric Sci 149(01):23–31. https://doi.org/10.1017/S0021859610000687

Debela N (2017) Adaptation to climate change in pastoral and agropastoral systems of Borana, southern Ethiopia. February, 168. https://api.semanticscholar.org/CorpusID:134567064

Dornan P, Jose M, Portela O, Pells K (2014) Climate shocks, food and nutrition security: evidence from the Young Lives cohort study. Oxfam Research Report, September, 56. https://doi.org/10.1164/rccm.201302-0365OC

Dynamics C, Berkes F (2006) From community-based resource management to complex systems: the scale issue and marine commons. https://www.researchgate.net/publication/42763830

Fadina R, Barjolle D (2018) Farmers' adaptation strategies to climate change and their implications in the Zou department of south Benin. https://doi.org/10.3390/environments5010015

Gebrehiwot T, Van Der Veen A (2013) Farm level adaptation to climate change: the case of farmer's in the Ethiopian highlands. Environ Manag 52(1):29–44. https://doi.org/10.1007/s00267-013-0039-3

Graham J, Rashid S, Malek M (2013) Disaster response and emergency risk management in Ethiopia. Food Agric Ethiop: Prog Policy Challenges 9780812208:256–279. https://doi.org/10.9783/9780812208610.256

Harvey B, Singh R (2017) Climate services for resilience: the changing roles of NGOs in Burkina Faso. https://doi.org/10.13140/RG.2.2.22999.37285

Hernandez J, Meisner J, Jacobs LA, Rabinowitz PM (2022) Re-centering indigenous knowledge in climate change discourse. PLOS Clim 1(5):e0000032. https://doi.org/10.1371/journal.pclm.0000032

IFAD IF for AD (2016) The traditional knowledge advantage. Family Farming Knowledge Platform https://www.ifad.org/documents/10180/2a1e3eb4-51a3-4746-8558-2fc1e6d3e645

Intergovernmental Panel on Climate Change (IPCC) (2014) Climate change 2014: Synthesis report. In: Contribution of Working Groups I, II, and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press. https://www.ipcc.ch/report/ar5/syr/

Jessen TD, Ban NC, Claxton NXEMŦOLTW, Darimont CT (2022) Contributions of indigenous knowledge to ecological and evolutionary understanding. Front Ecol Environ 20(2):93–101. https://doi.org/10.1002/fee.2435

Kebede W, Belay S (2017) The nexus between culture, indigenous knowledge and development in Ethiopia: review of existing literature. Glob J Hum Soc Sci 17(4):1–7. https://api.semanticscholar.org/CorpusID:202588953

Khalid MS, Slættalíð T, Parveen M, Hossain MS (2015) A systematic review and meta-analysis of teachers' development of digital literacy. Proceedings of the 1th D4|Learning International Conference Innovations in Digital Learning for Inclusion, November, 136–144. https://doi.org/10.13140/RG.2.1.2421.5120

Khatri I, Berkowska MA, van den Akker EB, Teodosio C, Reinders MJT, van Dongen JJM (2021) Population matched (PM) germline allelic variants of immunoglobulin (IG) loci: relevance in infectious diseases and vaccination studies in human populations. Genes Immun. 22:172–186. https://doi.org/10.1038/s41435-021-00143-7

Komba C, Muchapondw E (2015) Environment for development adaptation to climate change by smallholder farmers in Tanzania. https://www.jstor.org/stable/resrep15022

Koubi V (2019) Climate change and conflict. Annu Rev Polit Sci 22:343–360. https://doi.org/10.1146/annurev-polisci-050317-070830

Lewis M (2017) The power of the two. Retrieved from https://www.lewis.com/blog/the-power-of-the-two

Liehr S, Röhrig J, Mehring M, Kluge T (2017) How the social-ecological systems concept can guide transdisciplinary research and implementation: addressing water challenges in central northern Namibia. 1–19. https://doi.org/10.3390/su9071109

Mabhaudhi T, Chimonyo VGP, Hlahla S, Massawe F, Mayes S, Nhamo L, Modi AT (2019) Prospects of orphan crops in climate change. Planta 250(3):695–708. https://doi.org/10.1007/s00425-019-03129-y

Mafongoya PL, Ajayi OC (2017) Indigenous knowledge systems and climate change management in Africa. In Publications. Cta, Int (Issue November)

Magni G (2017) Indigenous knowledge and implications for the sustainable development agenda. Eur J Educ 52(4):437–447. https://doi.org/10.1111/ejed.12238

Makate C (2020) Local institutions and indigenous knowledge in adoption and scaling of climate-smart agricultural innovations among sub-Saharan smallholder farmers. Int J Clim Chang Strateg Manag 12(2):270–287. https://doi.org/10.1108/IJCCSM-07-2018-0055

Mane SN, Gadalkar SM, Rathod VK (2018) Intensification of paracetamol (acetaminophen) synthesis from hydroquinone using ultrasound. Ultrason Sonochem 49:244–251. https://www.researchgate.net/publication/326565575

Maru Y, Gebrekirstos A, Haile G (2019) Farmers' indigenous knowledge of tree conservation and acidic soil amendments: the role of "baabbo" and "Mona" systems: lessons from Gedeo community, southern Ethiopia. Cogent Food Agric 5(1):1–17. https://doi.org/10.1080/23311932.2019.1645259

Matewos T (2019a) Climate change-induced impacts on smallholder farmers in selected districts of Sidama, southern Ethiopia. Climate 7(5). https://doi.org/10.3390/cli7050070

Matewos T (2019b) Deconstructing institutional roles in climate change adaptation: the case of local public institutions in drought-prone districts of Sidama, southern Ethiopia. Environ Sci Policy 98(May):47–53. https://doi.org/10.1016/j.envsci.2019.05.005

Matewos T (2020) The state of local adaptive capacity to climate change in drought-prone districts of rural Sidama, southern Ethiopia. Clim Risk Manag 27(May 2019):100209. https://doi.org/10.1016/j.crm.2019.100209

Mclean KG, Nakashima D (2012) Weathering uncertainty: traditional knowledge for climate change assessment and adaptation: Policy commons. https://policycommons.net/artifacts/2102482/

Mekonnen A, Tessema A, Ganewo Z, Haile A (2021) Climate change impacts on household food security and farmers adaptation strategies. J Agric Food Res 6:100197. https://doi.org/10.1016/j.jafr.2021.100197

Mesfin D, Mirkuzie W, Ayinengida A, Fitsum (2020) Perceived organizational culture and its relationship with job satisfaction in primary hospitals of Jimma zone and Jimma town administration, correlational study. BMC Health Serv Res 20:438. https://doi.org/10.1186/s12913-020-05319-x

More YH, Madolo MC (2020) Effects of soil and water conservation structures on crop yield, Lemo District, Southern Ethiopia. Int J Sci Food Agric 4(4):519–527. https://doi.org/10.26855/ijfsa.2020.12.021

Mugambiwa SS (2018) Adaptation measures to sustain indigenous practices and the use of indigenous knowledge systems to adapt to climate change in Mutoko rural district of Zimbabwe. Jamba: J Disaster Risk Stud 10(1):1–9. https://doi.org/10.4102/jamba.v10i1.388

Mugambiwa SS (2021) Evoking the epistemology of climate governance through indigenous knowledge systems for sustainable development in rural Zimbabwe. Jamba: J Disaster Risk Stud 13(1):1–10. https://doi.org/10.4102/JAMBA.V13I1.1024

Mugambiwa SS, Makhubele JC (2021) Indigenous knowledge systems based climate governance in water and land resource management in rural Zimbabwe. J Water Clim Chang 12(5):2045–2054. https://doi.org/10.2166/wcc.2021.183

Muluken TK (2020) The role of indigenous conflict resolution mechanisms in the pastoral community: an implication for social solidarity in Somali region, Shineli Woreda. Open Access Library Journal 7:e6122. https://doi.org/10.4236/oalib.1106122

Mutambisi T, Chanza N, Matamanda AR, Ncube R, Chirisa I (2021) Climate change adaptation in Southern Africa: universalistic science or indigenous knowledge or hybrid. Afr Handb Clim Chang Adapt 1751–1766. https://doi.org/10.1007/978-3-030-45106-6_8

Muyambo CC, Thomas DSG (2016) The role of indigenous knowledge in drought risk reduction: A case of communal farmers in South Africa. Jamba: J Disaster Risk Stud 9(1):420. https://doi.org/10.4102/jamba.v9i1.420

Mwaniki F (2016) Kenyan farmers' perceptions of and adaptations to climate change before and after a radio program intervention; James Cook University; 2016. Jamba: Journal of Disaster Risk Studies 9(1). https://doi.org/10.4102/jamba.v9i1.420

Nyong A, Adesina F, Osman Elasha B (2007) The value of indigenous knowledge in climate change mitigation and adaptation strategies in the African Sahel. Mitig Adapt Strat Glob Chang 12(5):787–797. https://doi.org/10.1007/s11027-007-9099-0

Obi C, Bartolini F, D'Haese M (2021) Digitalization and migration: the role of social media and migrant networks in migration decisions. An exploratory study in Nigeria. Digit Policy Regul Gov 23(1):5–20. https://doi.org/10.1108/DPRG-08-2020-0101

Ochieng MD, Ahmed A (2014) The effects of privatization on the financial performance of Kenya Airways. J Bus Econ 5(10):1703–1716. https://api.semanticscholar.org/CorpusID:55778787

Ogalleh SA, Vogl CR, Eitzinger J, Hauser M (2012) Local perceptions and responses to climate change and variability: the case of Laikipia district, Kenya. 3302–3325. https://doi.org/10.3390/su4123302

Pearce LBT, Ford JD (2015) Systematic review approaches for climate change adaptation research. https://doi.org/10.1007/s10113-014-0708-7

Petros T (2010) Adoption of conservation tillage technologies in Metema Woreda, North Gondar zone, Ethiopia. M.Sc Thesis. Haromaya University

Rahut DB, Aryal JP, Marenya P ((2021)) Ex-ante adaptation strategies for climate challenges in Sub-Saharan Africa: macro and micro perspectives. Environ Challenges 3((December 2020)):100035. https://doi.org/10.1016/j.envc.2021.100035

Rahman HMT (2020) An analytical framework for assessing context-specific rural livelihood vulnerability. Sustainability 12:5654. https://doi.org/10.3390/su12145654

Reardon T, Delgado CL, Matlon PJ, Hopkins J (2001) Rural nonfarm employment and farming: Household-level linkages. The Electronic Journal of Agricultural and Development Economics 1(2):168–193. http://www.sciencedirect.com/science/article/pii/S0305-750X(00)00112-1

Reed MG, Henderson AE, Mendis-Millard S (2013) Shaping local context and outcomes: The role of governing agencies in collaborative natural resource management. Hum Dimens Wildl Int J 18(4):292–306. https://doi.org/10.1080/10871209.2013.801003

Regassa N, Stoecker BJ (2012) Household food insecurity and hunger among households in Sidama district, southern Ethiopia. Public Health Nutr 15(7):1276–1283. https://doi.org/10.1017/S1368980011003119

Reniko G, Mogomotsi PK, Mogomotsi GEJ (2018) Integration of indigenous knowledge systems in natural resources management in Hurungwe integration of indigenous knowledge systems in natural resources management in Hurungwe district. 6874. https://doi.org/10.1080/18186874.2018.1475869

Robison C (2020) Decolonizing climate discourse and legitimating indigenous wisdom: toward an ecosystemic episteme. Sustainability 12:5654. https://doi.org/10.3390/su12145654

Schlingmann A, Graham S, Benyei P, Corbera E, Martinez Sanesteban I, Marelle A, Solemany-Fard R, Reyes-García V (2021) Global patterns of adaptation to climate change by indigenous peoples and local communities. A systematic review. Curr Opin Environ Sustain 51:55–64. https://doi.org/10.1016/j.cosust.2021.03.002

Senbeta AF, Sime G, Struik P (2022) Enset farming system–a resilient, climate-robust production system in south and south-western Ethiopia. Cogent Food Agric 8(1). https://doi.org/10.1080/23311932.2022.2074626

Sharifian A, Fernández-Llamazares Á, Wario HT, Molnár Z, Cabeza M (2022) Dynamics of pastoral traditional ecological knowledge: a global state-of-the-art review. Ecol Soc 27(1). https://doi.org/10.5751/ES-12918-270114

Shiferaw A, Hurni H, Zeleke G (2013) The role of indigenous knowledge in land management for carbon sequestration and ecological services in southern Ethiopia. J Econ Sustain Dev 4(17):79–92

Shikur ZH (2020) Agricultural policies, agricultural production and rural households' welfare in Ethiopia. J Econ Struct 9(1). https://doi.org/10.1186/s40008-020-00228-y

Singh RB (2012) Climate change and food security. Improving crop productivity in sustainable agriculture, October, 1–22. https://doi.org/10.1002/9783527665334.ch1

Solomon E, Aaron K (2020) The impact of digital technology usage on economic growth in Africa. J Util Policy 67:101104–101017. https://api.semanticscholar.org/CorpusID:221371403

Songok CK, Kipkorir EC, Mugalavai EM (2011) Integration of indigenous knowledge systems into climate change adaptation and enhancing food security in Nandi and Keiyo districts, Kenya. In: Filho WL (ed) Experiences of climate change adaptation in Africa. Springer, Hamburg, pp 69–95

Šūmane S, Kunda I, Knickel K, Strauss A, Tisenkopfs T, des los Rios I, Rivera M, Chebach T, Ashkenazy A (2018) Local and farmers' knowledge matters! How integrating informal and formal knowledge enhances sustainable and resilient agriculture. J Rural Stud 59(232):241. https://doi.org/10.1016/j.jrurstud.2017.01.020

Tesfaye S, Guyassa E, Joseph Raj A, Birhane E, Wondim GT (2014) Land use and land cover change, and woody vegetation diversity in human driven landscape of Gilgel Tekeze catchment, northern Ethiopia. Int J For Res 2014:1–10. https://doi.org/10.1155/2014/614249

Theodory TF (2016) Dealing with change: indigenous knowledge & adaptation to CC in the Ngono River Basin, Tanzania. PhD.Tanzania, p 251

Tschakert P, Dietrich KA (2018) Anticipatory learning for climate change adaptation and resilience. Ecol Soc 15(2):11

United Nations Framework Convention on Climate Change (2013) UNFCCC NAP central. Retrieved January 15, 2022, from http://www4.unfccc.int/nap/Pages/Home.aspx

Upreti YG (1999) Indigenous knowledge, agricultural practices and food security in developing countries: opportunities and challenges. https://www.academia.edu/1243525/

Van Huynh C, Le Phuong QN, Hong Nguyen MT, Tran PT, Nguyen TQ, Pham TG, Khanh Nguyen LH, Dieu Nguyen LT, Trinh HN (2020) Indigenous knowledge in relation to climate change: adaptation practices used by the Xo Dang people of central Vietnam. Heliyon 6(12):e05656. https://doi.org/10.1016/j.heliyon.2020.e05656

Vinke K, Rottmann S, Gornott C, Zabre P, NaynaSchwerdtle P, Sauerborn R (2022) Is migration an effective adaptation to climate-related agricultural distress in Sub-Saharan Africa? Popul Environ 43(3):319–345. https://doi.org/10.1007/s11111-021-00393-7

Wondim A, Genenew A, Abaye G, Demissie C (2018) Indigenous knowledge adaption strategies in response to climate change: the case of North Gondar, Ethiopia. J Earth Sci Clim Chang 09(05). https://doi.org/10.4172/2157-7617.1000470

Xu L, Gober P, Wheater HS, Kajikawa Y (2018) Reframing socio-hydrological research to include a social science perspective. J Hydrol 563:76–83. https://doi.org/10.1016/j.jhydrol.2018.05.061

Yemataw Z, Mekonen A, Chala A, Tesfaye K, Mekonen K, Studholme DJ, Sharma K (2017) Farmers' knowledge and perception of enset Xanthomonas wilt in southern Ethiopia. Agric Food Secur. November, 0–12. https://doi.org/10.1186/s40066-017-0146-0

Yigezu Wendimu G (2021) The challenges and prospects of Ethiopian agriculture. Cogent Food Agric 7(1). https://doi.org/10.1080/23311932.2021.1923619

Yona Y, Mathewos T (2017) Assessing challenges of non-farm livelihood diversification in Boricha Woreda, Sidama zone. J Dev Agric Econ 9(4):87–96. https://doi.org/10.5897/JDAE2016.0788

Yua E, Raymond-Yakoubian J, Daniel RA, Behe C (2022) A framework for co-production of knowledge in the context of Arctic research. Ecol Soc 27(1):34. https://doi.org/10.5751/ES-12960-270134

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.